PhishLens

Design and Development of an Explainable Browser Extension for Real-Time Phishing Website Detection

Approbation-period submission | 24 October 2026

Software Engineering • Group SE-2422

Team member	GitHub account	Planned role
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Submission scope

This report consolidates the team's requirements and planning evidence, a preliminary university-student survey, a literature and technology review, design artefacts, an interface mock-up, an evaluation plan and the supervisor's supplied feedback. The existing project scope and task dates are retained.

Stage: analysis and design. No trained model, deployable browser extension, backend service, detection result or measured performance is claimed. The report was assembled with AI assistance; individual authorship and review must be established through the team's actual work records.

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University submission: 24 October 2026, 18:00, Asia/Almaty. PHD-1 task deadline: 26 October 2026. These dates refer to different milestones.

1. Proposal and scope

Problem and audience

University students and other everyday internet users encounter suspicious links and may find a warning difficult to interpret. The project investigates whether a browser extension can combine a risk estimate with understandable, evidence-linked reasons. The current research sample consists of adults who self-identified as university students; it is not a school-pupil study.

Aim and objectives

The aim is to design and develop an explainable browser extension for real-time phishing risk assessment. Objectives are to investigate user needs and related work; select reproducible data and URL features; compare baseline models; develop a Chromium extension and minimal API contract; present risk and reasons; and evaluate detection quality, runtime, reliability and usability.

First working version

The first MVP preserves the supplied scope: active-page URL checking, a baseline model, local inference where feasible, a popup with risk and reasons, and a minimal FastAPI service. Later stages add DOM/domain features, explanation packaging, reputation checks and caching, settings and kk/ru/en localization, feedback storage and broader evaluation.

A risk indication must not promise that a site is safe. An unavailable analysis must remain unavailable; it must not be converted into a low-risk result. Local processing is preferred. Browsing history, form values, passwords, cookies and page content are not uploaded by default.

Expected deliverables

The diploma deliverables are an installable extension, documented supporting components, reproducible model/data procedures, a comparison of alternatives and a report on quality, latency, usability and limitations. At this checkpoint, the deliverables are the design and evidence package and an interactive UI mock-up.

Contribution and novelty to investigate

The intended contribution is a student-oriented workflow with inspectable reasons and a local-first design. This is a design proposition, not a proven scientific novelty or superiority claim. Existing browser protections and research detectors already address phishing. The proposed workflow must demonstrate value through comparative evaluation.

2. User evidence — method and findings

Source: seven Google Forms summary screenshots supplied by the team, captured on 24 October 2026 around 17:19. They show 12 submitted responses to an English multiple-choice questionnaire. Counts below are reconstructed from displayed percentages and denominators; the raw response export was not supplied.

All 12 responses selected consent confirming adulthood and university-student status. Respondent identity, unique participation, recruitment coverage and cross-question combinations cannot be verified from the screenshots. Treat this as an exploratory convenience sample, not a representative estimate.

Question	Observed results
Q1 Consent (n=12)	12 consenting responses (100%).
Q2 Study level (n=12)	Year 2: 4 (33.3%); year 3: 8 (66.7%); other levels: 0.
Q3 Main browser (n=12)	Chrome: 6 (50%); Edge: 2 (16.7%); Safari: 3 (25%); another browser: 1 (8.3%).
Q4 Phishing familiarity (n=12)	Never heard: 1; heard but cannot explain: 1; general understanding: 7 (58.3%); understands well and can name signs: 2; unsure: 1.
Q5 Suspicious links in 3 months (n=12)	Never: 2; 1–2 times: 4; 3–5 times: 5; more than 5: 0; cannot remember: 1.
Q6 Checks before sensitive action (multiple choices; n=12)	Address spelling: 4; find official site: 4; HTTPS: 2; reviews/reputation: 3; browser/antivirus warnings: 4; usually nothing: 1; avoid entering information: 0.

Interpretation for scope

Eight responses use Chrome or Edge, supporting a Chromium-first prototype as a practical starting point; the remaining four responses make later compatibility a relevant issue. Nine responses reported at least one suspicious encounter, while only two claimed strong familiarity with warning signs. These findings motivate an understandable warning workflow without proving the extension will improve behaviour.

Q6 is multiple-select; totals can exceed 100%. HTTPS is a reported user habit, not proof of website legitimacy. Counts are retained as supplied; no respondent-level records have been invented.

2. User evidence — decisions and limitations

Question	Observed results
Q7 Warning preference (n=11)	Short warning: 3 (27.3%); specific reason and action: 1 (9.1%); numeric risk: 3 (27.3%); detailed technical description: 4 (36.4%). One response is missing.
Q8 Data-processing preference (n=12)	Local only: 6 (50%); external checks per website consent: 2; one-time consent: 3; would not use: 0; unsure: 1.
Q9 Last action (n=12)	Did not open/closed: 3; checked address: 2; antivirus/link checker: 1; asked someone: 2; continued: 2; other: 2.
Q10 Warning language (n=12)	Russian: 5; Kazakh: 2; English: 2; another language: 1; no preference: 2.
Q11 Interest in the idea (n=12)	Definitely would try: 6; probably would try: 2; unsure: 2; probably would not: 1; definitely would not: 1.
Q12 Barriers (multiple choices; n=12)	Slowdown: 0; incorrect warnings: 1; privacy: 0; setup: 2; unclear explanations: 1; incompatibility: 2; no need: 1; other: 0; none of above: 9; unsure: 2.

Design decisions derived from this sample

Keep the first version on Chromium (Q3, FR-06). Use a brief result with expandable technical reasons (Q7, FR-03); no single warning format has a majority. Prefer local checking and require explicit external-check consent (Q8, FR-05). Preserve later kk/ru/en localization (Q10). Eight of twelve expressed interest in trying the idea (Q11); this is interest, not a usability test or demonstrated effectiveness.

Data quality and remaining evidence

Q12 contains inconsistent exclusive-option selections: nine “none” and two “unsure” responses would leave only one response for substantive barriers if exclusivity were respected, yet setup and incompatibility each have two selections. Preserve these raw aggregates and do not rank barriers confidently until the individual export is checked. No records were removed or repaired from summary charts.

The supplied evidence does not include records of 3–5 user discussions or a substantive agreement with a client. The survey provides preliminary user evidence but does not by itself establish that this separate checkpoint requirement has been satisfied. Supervisor feedback below concerns the materials and packaging; it is not treated as a client interview.

3. Literature review — phishing detection

[S01] Lin Y. et al. (2021). Phishpedia: A Hybrid Deep Learning Based Approach to Visually Identify Phishing Webpages. USENIX Security, 3793–3810.

Phishpedia identifies brand impersonation through logo recognition and domain comparison. This makes a brand mismatch explainable. A maintained reference set and visual inference are required; the published evaluation does not establish performance on our browser or local audience. We retain the idea of linking warnings to observable evidence, while excluding a full visual pipeline from the first MVP.

[S02] Liu R. et al. (2022). Inferring Phishing Intention via Webpage Appearance and Dynamics: A Deep Vision Based Approach. USENIX Security, 1633–1650.

PhishIntention combines apparent brand identity with credential-taking intent and page interaction. It shows why visual similarity alone may produce misleading alerts. Dynamic interaction adds cost and complexity. For PhishLens, later DOM analysis should examine structure without collecting form values or interacting with live phishing pages.

[S03] Liu R. et al. (2023). Knowledge Expansion and Counterfactual Interaction for Reference-Based Phishing Detection. USENIX Security, 4139–4156.

DynaPhish expands reference knowledge and explores pages that lack familiar branding. This addresses coverage limits of fixed reference lists, but maintaining and validating the knowledge base becomes part of the system. Our evaluation should include unfamiliar brands; absence of a known logo cannot be treated as evidence of safety.

[S04] Liu R. et al. (2024). Less Defined Knowledge and More True Alarms: Reference-based Phishing Detection without a Pre-defined Reference List. USENIX Security, 523–540.

PhishLLM uses language-model knowledge and validation to reduce dependence on a predefined brand list. This offers broad brand coverage but introduces model reliability and external-service dependencies. We exclude an LLM from the mandatory local inference path and will express reasons from measured features instead of generating unverified accusations.

3. Literature review — robustness and explanations

[S05] Li Y. et al. (2024). KnowPhish: Large Language Models Meet Multimodal Knowledge Graphs for Enhancing Reference-Based Phishing Detection. USENIX Security, 793–810.

KnowPhish combines multimodal brand knowledge with textual and visual information, including cases where logos alone are insufficient. The knowledge infrastructure exceeds the first MVP. The transferable lesson is to evaluate coverage across site types and brands rather than assume one visual cue is universal.

[S06] Ji F. et al. (2025). Evaluating the Effectiveness and Robustness of Visual Similarity-based Phishing Detection Models. USENIX Security, 3201–3220.

The authors examine real-world effectiveness and robustness of visual similarity detectors, including evasion through visual changes. Their findings concern the evaluated visual approaches, not every phishing model. They motivate independent testing, distribution-shift checks and explicit error analysis rather than reporting a single favourable random-split score.

[S07] Ribeiro M. T., Singh S., Guestrin C. (2016). “Why Should I Trust You?”: Explaining the Predictions of Any Classifier. KDD, 1135–1144.

LIME approximates a model locally with an interpretable explanation. A local approximation can help inspect a prediction, but its stability and perturbation assumptions must be checked. We will compare explanation consistency and runtime; a model explanation is not proof that a website is malicious.

[S08] Lundberg S. M., Lee S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 30.

SHAP assigns feature contributions to model predictions. This supports analysis of which URL features influence a score, subject to the model and explanation assumptions. Contributions do not establish causality. We will check whether a compact set of contributions can be translated faithfully into user-facing reasons within the latency budget.

S01–S06 provide a recent 2021–2025 phishing-research core. S07–S08 are foundational explanation methods retained for their methodological role. The review compares published approaches; their performance numbers are not presented as project results.

3. Literature synthesis and data choice

Synthesis

The reviewed work moves from identifying brand impersonation toward intent, expanding reference coverage and handling robustness in realistic conditions. These approaches can offer richer evidence than URL-only classification, but add data, computation and maintenance requirements. PhishLens therefore starts with a reproducible URL baseline and treats richer features as later scope. A simpler baseline is easier to reproduce, but may miss attacks that require page context.

Explanations must distinguish a measured feature from the model's interpretation. A long or unusual URL is an indicator, not proof of malicious intent. The interface should communicate uncertainty and permit details rather than convert a score into a definitive accusation. SHAP/LIME will be evaluated for fidelity, stability and cost; explanation quality must be tested separately from detection quality.

Candidate data and leakage controls [S09]

Candidate dataset: 235,795 instances and 54 features, with 134,850 legitimate and 100,945 phishing cases. The original label is 1 for legitimate and 0 for phishing; evaluation will use phishing_positive = 1 - label. The CC BY 4.0 attribution must be retained. URL and page-derived features must be separated; dataset availability does not establish suitability for our deployment.

The first dataset pipeline will document collection context, licence, label mapping and selected feature definitions. Remove exact duplicates and prevent the same registrable domain from leaking across training and final evaluation where feasible. Fit preprocessing and learned statistics only on training data. Use an untouched final test partition. Report any inability to create a genuinely time-separated sample.

Do not use all precomputed columns blindly: some require page content or knowledge unavailable during URL-only inference. Exclude filenames, identifiers, labels and columns that leak target information. Confirm identical feature order and normalization in training and the extension. Record dataset/model checksums, random seeds and dependency versions.

Engineering sources [S10–S15]

Safe Browsing, SmartScreen and Netcraft inform the system comparison and optional reputation design. Chrome MV3 documentation constrains the extension lifecycle. ONNX Runtime Web provides a candidate browser inference path, whose model and execution compatibility remain to be tested. FastAPI supports validated, documented contracts. None of these documents proves the proposed system's accuracy, privacy implementation or latency.

4. Comparison of existing systems

Three existing protections are compared using eight criteria. This is a documented feature comparison, not a controlled accuracy benchmark. “Not specified” means the referenced public description does not establish that property. Sources: [S10–S12].

Criterion	Google Safe Browsing	Edge SmartScreen	Netcraft extension
Delivery	API/platform protection; host application presents UI	Built into Edge	Browser extension
Detection signals	Unsafe-resource lists	Reputation and page analysis	Threat intelligence and site information
User output	Threat matches for the integrating product	Browser warning	Warning and site information
Explanation detail	API threat type; application controls wording	Warning; model internals not specified	Site/context information; model internals not specified
External dependency	Threat-list/reputation infrastructure	Microsoft protection services	Netcraft protection services
Browser coverage	Depends on integration	Edge integration	Chrome, Firefox, Edge and Opera listed
Reporting / feedback	Depends on host integration	User warning/feedback workflow	Suspicious-site reporting
Local feature attribution	Not specified by API overview	Not specified by public overview	Not specified by product overview

Implications for PhishLens

PhishLens should complement browser protections with inspectable project-defined indicators, explicit unavailable states and a documented data flow. It must not claim to outperform existing systems without a common labelled test protocol. Reputation checks are an optional later signal; no match must remain inconclusive.

The proposed system differs in research emphasis, not proven superiority: URL-feature explanations and a student-focused evaluation are explicit design goals. Browser-native protection remains enabled during tests. Safe Browsing deployment terms must be checked before use; a production reputation integration is not part of this submission.

5. Technology decisions

These are proposed engineering choices for the preserved MVP. Each decision includes alternatives and a criterion for revisiting it; none is an implemented benchmark result.

Decision	Alternatives and trade-off	Proposed choice / validation
Extension UI	Vanilla TypeScript: small explicit state model. React: reusable components, more tooling. Vue: component structure with framework dependency.	TypeScript with simple UI first; adopt a framework only if UI complexity warrants it. Test popup lifecycle and accessible state changes.
Model baseline	Logistic regression: transparent linear baseline. Random forest: non-linear tabular model. Gradient boosting: additional capacity and tuning.	Compare all three with the same split and URL-only features. Choose on recall, false positives, F1, size and runtime rather than accuracy alone.
Inference location	Local ONNX: avoids default URL uploads, requires export/operator support. JavaScript rules: simple baseline, limited adaptability. Server inference: easier deployment but adds network/privacy dependency.	Try local ONNX [S14], retain rules as comparison. Do not silently fall back to server upload. Export/CSP/runtime feasibility is a gate.
API framework	FastAPI: Python schemas/OpenAPI [S15]. Django: broader web stack. Express: JavaScript ecosystem but separate ML runtime integration.	FastAPI, matching the existing scope. Verify request validation, timeout and explicit error contracts.
Persistence, later stage	PostgreSQL: relational constraints and multiuser service. SQLite: simple local/prototype storage. MongoDB: flexible document structure.	Start without server browsing-history storage. Choose PostgreSQL only when approved feedback/result entities require a shared service; define retention first.
Communication	REST: explicit request/response. WebSocket: persistent events, more lifecycle work. gRPC: typed service contracts but browser adaptation.	REST for occasional checks. Separate optional service latency from local inference timing; document versioned schema.

Supporting planned tools: GitHub for version control and review, YouTrack for tasks, Docker and CI in later planned tasks. Redis is deferred until a measured caching requirement exists. Framework choice does not replace security or performance testing.

6. Functional requirements

The supplied PHD-1 requirements are preserved below. FR-01–FR-06 are Must-have for the first working version; richer features remain later scope.

ID	Requirement and acceptance condition
FR-01	Check the active page URL on supported navigation events. Ignore internal browser pages and handle missing or invalid URLs without crashing.
FR-02	Apply a reproducible feature pipeline and baseline model. Return a documented risk result; show unavailable or uncertain results explicitly.
FR-03	Display the result in the popup with understandable reasons linked to observable indicators. Present the result as a risk assessment, with limitations.
FR-04	Define the extension and ML input/output contract. Provide a FastAPI service with validated schemas, documented responses, and clear failure handling.
FR-05	Keep browsing history off the server. Document any individual URL or domain sent for a selected reputation check and review this data flow before enabling it.
FR-06	Make the extension installable in developer mode from the repository. Document installation, startup, a safe demonstration, and known limitations.

Traceability from evidence

FR-03 responds to varied warning preferences in Q7 and the explanation literature [S07–S08]. FR-05 responds to Q8 local-processing preference and the documented external-service boundary. FR-06 reflects Q3 Chromium usage. FR-01 and FR-02 establish the minimum measurable analysis workflow; the survey does not validate the classifier.

Safe behaviour for failures

An unsupported browser URL, a missing model, an invalid response or a timeout must produce an explicit unsupported/unavailable/error state. No failure path may display a “safe” result. Demonstrations and usability tasks use inert examples, without asking participants to open live malicious websites.

6. Nonfunctional targets and acceptance

ID	Preserved target and assessment
NFR-01	Detection quality: proposed F1 target of at least 0.90 on held-out data. Report precision, recall, F1, ROC-AUC, class balance, and errors. Split and deduplicate data to reduce leakage; do not tune on the final test set.
NFR-02	Latency: proposed local feature extraction plus inference target of at most 200 ms. Record browser, hardware, model, and measurement method. Measure external lookup/API latency separately and discuss the end-to-end result.
NFR-03	Reliability and usability: show loading, unavailable, timeout, and error states. Test safe examples without requiring users to open live phishing sites. Evaluate multilingual text and usability in the later system evaluation.

Operational detail added for future evaluation

For NFR-01, select thresholds on validation data and freeze them before final testing. Report confusion matrices and false-positive/false-negative examples, with phishing treated as the positive class. The F1 ≥ 0.90 threshold is an aspiration from the supplied scope, not an achieved score.

For NFR-02, report median and p95 local feature-extraction-plus-inference time over at least 100 repeated safe-fixture checks, with cold and warm runs separated. Use a documented machine/browser/model version. The proposed acceptance interpretation is p95 ≤ 200 ms; the supervisor has not explicitly approved this interpretation. Measure API, reputation lookup and total user-visible time separately.

For NFR-03, require visible loading, result, unavailable and error states. Test keyboard interaction and explanatory text. Later language evaluation should use human-reviewed Kazakh/Russian/English translations. An English submission does not require all product users to prefer English.

Privacy acceptance

Inspect network requests in local-only mode: no URL, domain, page content, form value, password, cookie or browsing history should leave the device. Optional external checks require separate consent and a documented payload. Any later feedback storage needs justified fields, retention and deletion. This is a testable design requirement, not a claim that an unbuilt system already meets it.

7. Architecture and interfaces

Context and use cases

The student initiates or receives an active-page check and reads a result. The extension owns the user interaction and preferred local analysis. The offline model-development pipeline supplies a versioned model and feature schema. A separately enabled API/reputation path is optional and crosses a data boundary.

View	Elements and relationships
Context	Student extension; model-build pipeline versioned model/schema extension; extension optional consented API later reputation provider.
Use cases	Check current page; read risk; expand reasons; handle unavailable result and retry; later change language and external-check consent.
Components	Popup event coordinator URL feature extractor model adapter result/explanation mapper popup. API adapter is an optional separate branch.
Data boundary	Default analysis remains local. No full page, form values, credentials or browsing history are sent. A later external check documents the exact selected URL/domain payload.

Proposed result contract

A versioned result contains status, risk category, modelVersion, featureSchemaVersion, reason codes and measured duration. A numerical score is optional and must be labelled as a model score unless calibrated probability is demonstrated. Error responses include an error code and a retry indication, not a fabricated risk.

Illustrative statuses: loading, analysed, unsupported, unavailable and error. Reason codes map to deterministic UI text, such as an unusual URL structure. Raw user-provided strings must be safely rendered. Model updates must preserve schema compatibility.

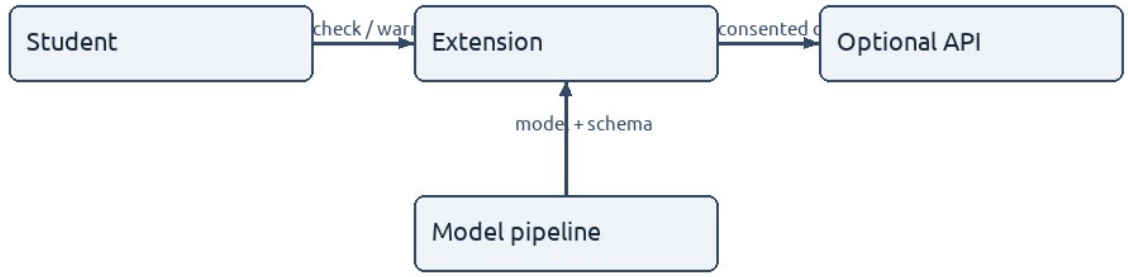
Editable diagrams

The repository includes context.mmd, use-cases.mmd and components.mmd with directional Mermaid relationships, plus corresponding SVG design views. They represent a proposed architecture. Local model export and service-worker/runtime compatibility require implementation tests [S13–S14].

7. Architecture — diagrams

Proposed relationships for the baseline and optional paths. Editable SVG and Mermaid sources are included in the repository.

System context



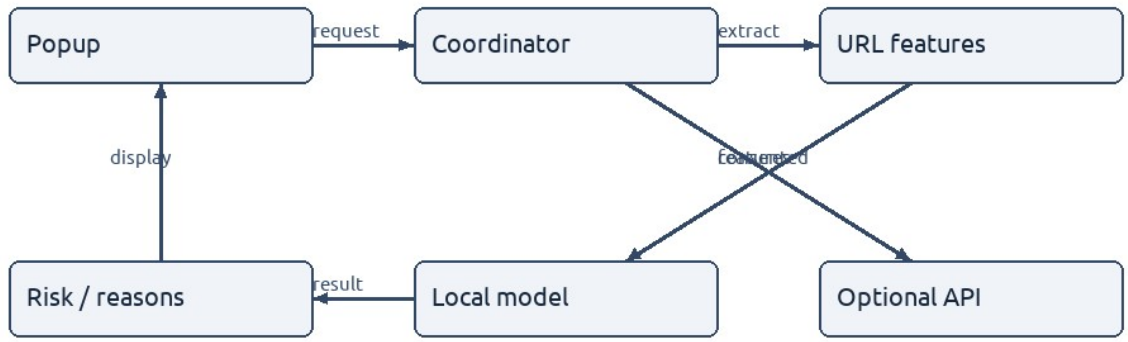
Context: the extension mediates student interaction; external requests are optional.

Use-case view



Use cases: checking, understanding results, retrying and later preferences.

Component view



Components: features and inference feed the result mapper; the API is optional.

8. Interface prototype

The supplied package includes an English interactive mock-up at prototype/index.html. Open the file in a browser; it requires no server and sends no network requests. It presents manually selected demonstration states and does not inspect the current website or classify a URL.

State	Visible behaviour	Design purpose
Loading	“Checking this example…”	Makes processing visible; no premature verdict.
Low risk	No strong warning signs in this demo; limitations remain visible.	Avoids declaring a website safe.
Warning	Risk warning with an example URL-structure reason; expandable detail.	Supports a brief result and optional technical explanation.
Unavailable	Analysis unavailable; retry demo control.	Separates missing evidence from low risk.

Warning hierarchy

The first line conveys the risk category. The next line states an observable reason; a Details control exposes technical context. The user can close the suspicious page independently. A warning should not force the user to trust a numerical score. The mock-up labels example data and future settings explicitly.

Survey connection

Q7 split preferences across detailed descriptions, scores and brief messages; a layered interface is a design response to that variation. Q8 supports local-only defaults. Q10 motivates later localization. No participant has been shown this final mock-up in the supplied evidence, so no usability success claim is made.

Prototype review checklist

Check all four states, Details expansion, the retry control, keyboard focus, narrow-screen layout and the fact that no network request occurs. These are interface checks only. They do not test phishing detection, model quality or browser-extension permissions.

9. Evaluation plan

Area	Planned protocol	Recorded outputs
Detection quality	Compare logistic regression, random forest and gradient boosting on a shared leakage-controlled split. Freeze thresholds and use an untouched test partition.	Precision, recall, F1, ROC-AUC, confusion matrix, class balance and confidence intervals where appropriate.
Baseline comparison	Compare the selected model against a simple documented URL-rule baseline. Where feasible evaluate reputation as a separate optional signal on the same cases.	Error types, added value, false positives and dependency coverage. Do not transfer published benchmark scores.
Performance	At least 100 repeated safe checks; separate cold/warm, local and network durations. Record browser, hardware, runtime and model.	Median, p95, sample count, setup and failures; proposed local p95 target ≤200 ms.
Reliability and privacy	Exercise unsupported pages, invalid URLs, missing models, service errors and consent states. Inspect network behaviour.	Expected/actual outcomes; no false safe result; documented request payloads.
Explanation quality	Compare reason text with underlying feature values/contributions and repeated runs. Test whether users identify the reason and recommended response.	Fidelity checks, stability observations, comprehension and misleading wording.
Later usability	Recruit target users, obtain consent and run fixed safe scenarios on the working prototype. Record completion, time, errors and comments.	Task results and anonymised observations; distinguish them from the current idea-interest survey.

Dataset and safety limitations

Static datasets may not represent new campaigns, local brands or adversarial changes. Domain grouping and a time-separated sample, when available, strengthen evaluation but cannot guarantee future detection. Safe archived/inert fixtures should be used instead of participant interaction with live phishing sites.

No evaluation in this section is reported as already completed. The current survey asks about habits, preferences and interest in an idea; it is not a detection benchmark or user test of a finished extension.

10. Team, workflow and evidence

Member / GitHub	Planned responsibilities
Aiym Pangerey / aiympangerey1703	Team coordination; architecture, URL workflow, API and integration. Frontend/backend shared with Sabina.
Sabina Zhumagaliyeva / 6wq22pbr26-droid	Popup/warnings, settings, localization, storage, system and usability testing. Frontend/backend shared with Aiym.
Alibek Asylbekuly / alibek44	Dataset/features, model training and evaluation, explanation and packaging. Joint integration verification.

Version control and task process

Use a branch with the real PHD task ID, a coherent commit, a pull request and another member's actual review. Link the PR/commit from YouTrack. Keep work-in-progress status until acceptance criteria are met. A prepared document is not evidence of three independent contributions; no commits or reviews on behalf of other members are asserted here.

The repository is private. Earlier setup recorded invitations to Aiym and Alibek and an English README pull request. Current acceptance, reviewer access, merge status and the final package upload must be verified in GitHub before submission. No unverified acceptance or upload is claimed in this report.

YouTrack snapshot

The supplied board is Explainable Phishing Detection, project PHD. It shows 19 cards, with 18 in Backlog and PHD-1 in Review; none are shown Done. The PHD-1 image contains three attached review documents. It predates the newly supplied supervisor reply, so the screenshot is retained unchanged as historical evidence.

Some ML tasks display the assignee "admin". The role plan assigns ML work to Alibek, but the screenshots do not prove that admin is his account. This mapping should be checked. PHD-16 remains assigned to Aiym as shown. Dates and completed work have not been rewritten to make progress appear greater.

Repository: https://github.com/6wq22pbr26-droid/phishlens-diploma
YouTrack instance: https://aiympangerey1703.youtrack.cloud

The exact project/board URL and committee access must be copied/verified from the live service. An instance URL alone does not prove access to a private board.

10. Milestones and dependencies

Dates below preserve the supplied team schedule. English labels are shortened for readability. They are planned deliverables, not records of completion. Overall planned project end: 10 May 2027.

Task	Deliverable	Planned owner	Due
PHD-1	Requirements and MVP scope	Aiym	2026-10-26
PHD-2	Architecture and backend structure	Aiym	2026-11-13
PHD-3	Manifest V3 and URL checking	Aiym	2026-12-11
PHD-4	Phishing analysis REST API	Aiym	2026-12-25
PHD-5	ML model integration	Aiym	2027-03-26
PHD-6	Popup and warning UI	Sabina	2026-12-18
PHD-7	Settings and localization	Sabina	2027-02-12
PHD-8	Extension/API integration	Sabina	2027-01-15
PHD-9	Results and feedback storage	Sabina	2027-01-22
PHD-10	Functional, load and usability testing	Sabina	2027-04-23
PHD-11	Research and dataset selection	Alibek*	2026-11-06
PHD-12	Data preparation and URL features	Alibek*	2026-11-27
PHD-13	Baseline model training	Alibek*	2027-01-15
PHD-14	Model evaluation	Alibek*	2027-03-12
PHD-15	Explanations and model packaging	Alibek*	2027-02-26
PHD-16	Reputation integration	Aiym	2027-02-05
PHD-17	Caching	Team / verify assignee	2027-02-19
PHD-18	Docker packaging	Team / verify assignee	2027-01-29
PHD-19	Continuous integration	Team / verify assignee	2026-12-04

* ML tasks are planned for Alibek in the role document, while the supplied screenshot shows “admin”; verify identity. Assignees for PHD-17–PHD-19 are not established by the supplied visible evidence.

Dependencies: requirements architecture; dataset/features training evaluation and packaging; API contract + extension shell integration; integrated system system/usability testing. CI may begin before implementation and remain active throughout. The 26 October PHD-1 date is a task-planning date, while the university report is due on 24 October.

10. Risks and checkpoint readiness

Risk	Effect	Response
Small survey and missing discussions	User needs may be weakly supported.	Keep conclusions exploratory; obtain genuine discussion records or a substantive client agreement.
Data leakage / unavailable online features	Optimistic scores that do not reproduce in the extension.	Group/deduplicate data, freeze test set and validate URL-only feature availability.
Model export or runtime incompatibility	Local inference target may fail.	Test a small exported baseline early; report failure explicitly instead of silently uploading data.
False positives / misleading reasons	Users may ignore warnings or trust an unsupported verdict.	Evaluate errors and explanation fidelity; avoid safe guarantees.
Private links / unresolved invitations	Commission may be unable to assess evidence.	Verify supervisor and commission access before form submission.
Task identity / contribution gaps	Responsibility and completed work are unclear.	Map admin to the real member; record actual work, PRs and review without backdating.

Evidence map

Requirement area	Included evidence / status
Proposal, literature, comparisons	Sections 1, 3–5; 15 sources including 8 scientific works.
Requirements, architecture, prototype, evaluation	Sections 6–9; editable scope, diagram sources and interactive mock-up.
User input	12 survey responses analysed; 3–5 discussion records or client agreement not supplied.
Planning and team	Role plan; 19-task schedule; authentic board/task screenshots.
Supervisor review	General positive feedback and packaging instruction supplied on 24 October; detailed technical review not documented.
Repository and access	Local package prepared; live upload, memberships, reviewer access and contribution evidence require verification.

11. Supervisor feedback and changes

Evidence: Telegram screenshot supplied by the team on 24 October 2026. The requirements document and board were shared around 16:59–17:00 and the updated role justification around 17:07. The contact displayed as Ayvar replied positively at 17:15 and gave submission-format guidance at 17:17. The team identifies this contact as the supervisor.

Feedback	Team response / change
General response: “Excellent!” (translation of the supplied Russian reply).	Recorded as general positive feedback on the sent materials. It is not interpreted as a signed approval, detailed validation of every technical target, or confirmation that all school requirements are met.
Convert the document to PDF; include the materials in the report. Materials may also be placed in a GitHub documentation folder.	Prepared one consolidated English PDF report including the survey, design sections and evidence appendices. Prepared a separate, readable PDF of YouTrack screenshots and a repository docs package.
No detailed requested technical changes are visible.	Preserved the existing MVP, roles and task dates. Corrected residual Russian names, integrated the real survey evidence and identified outstanding evidence explicitly.

Changes made for this submission

English team names and explanatory text; analysis of the supplied 12-response survey with Q7 denominator 11 and Q12 quality limitation; source-based literature and alternatives; traceable requirements and proposed tests; English mock-up and editable diagrams; a consolidated report and one separate board/task evidence PDF.

Changes deliberately not represented as completed

No interviews, client discussions, trained models, extension implementation, accuracy results, runtime measurements, teammate reviews or accepted invitations have been invented. The historical board images and the 26 October PHD-1 deadline remain unchanged. The reply does not remove the school’s separate evidence or access requirements.

The source screenshot is reproduced in Appendix C. The editable feedback and change-record documents use the same limited interpretation.

References

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[S07] Ribeiro M. T., Singh S., Guestrin C. (2016). “Why Should I Trust You?”: Explaining the Predictions of Any Classifier. KDD, 1135–1144.

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[S09] Prasad A., Chandra S. (2024). PhiUSIIL Phishing URL (Website). UCI Machine Learning Repository.

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[S10] Google. Safe Browsing API documentation.

https://developers.google.com/safe-browsing

[S11] Microsoft. Microsoft Edge support for Microsoft Defender SmartScreen.

https://learn.microsoft.com/en-us/deployedge/microsoft-edge-security-smartscreen

[S12] Netcraft. Browser Extension.

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[S13] Google. Extension service worker basics. Chrome for Developers.

https://developer.chrome.com/docs/extensions/develop/concepts/service-workers/basics

[S14] Microsoft. ONNX Runtime Web tutorials.

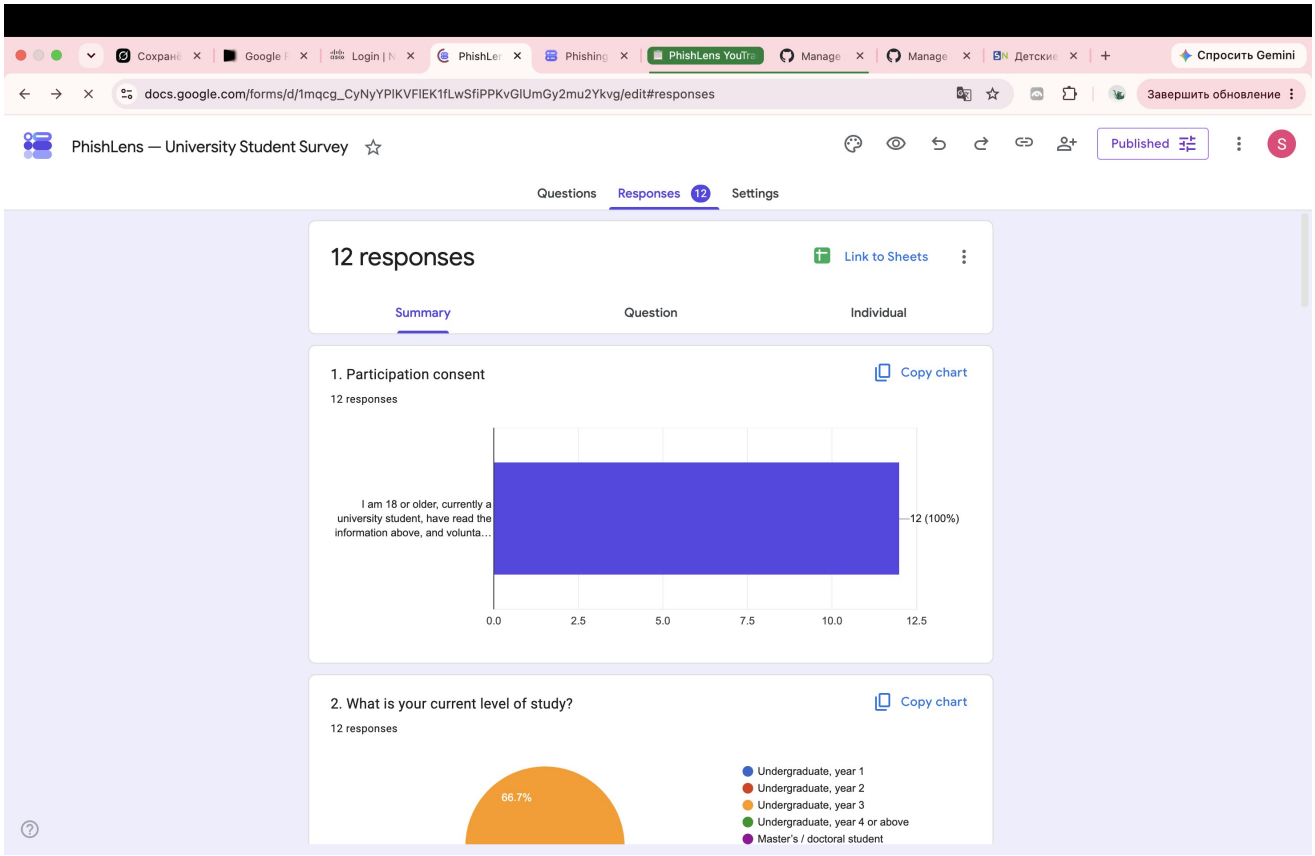
https://onnxruntime.ai/docs/tutorials/web/

[S15] FastAPI. Features.

https://fastapi.tiangolo.com/features/

Appendix A.1 — Survey evidence

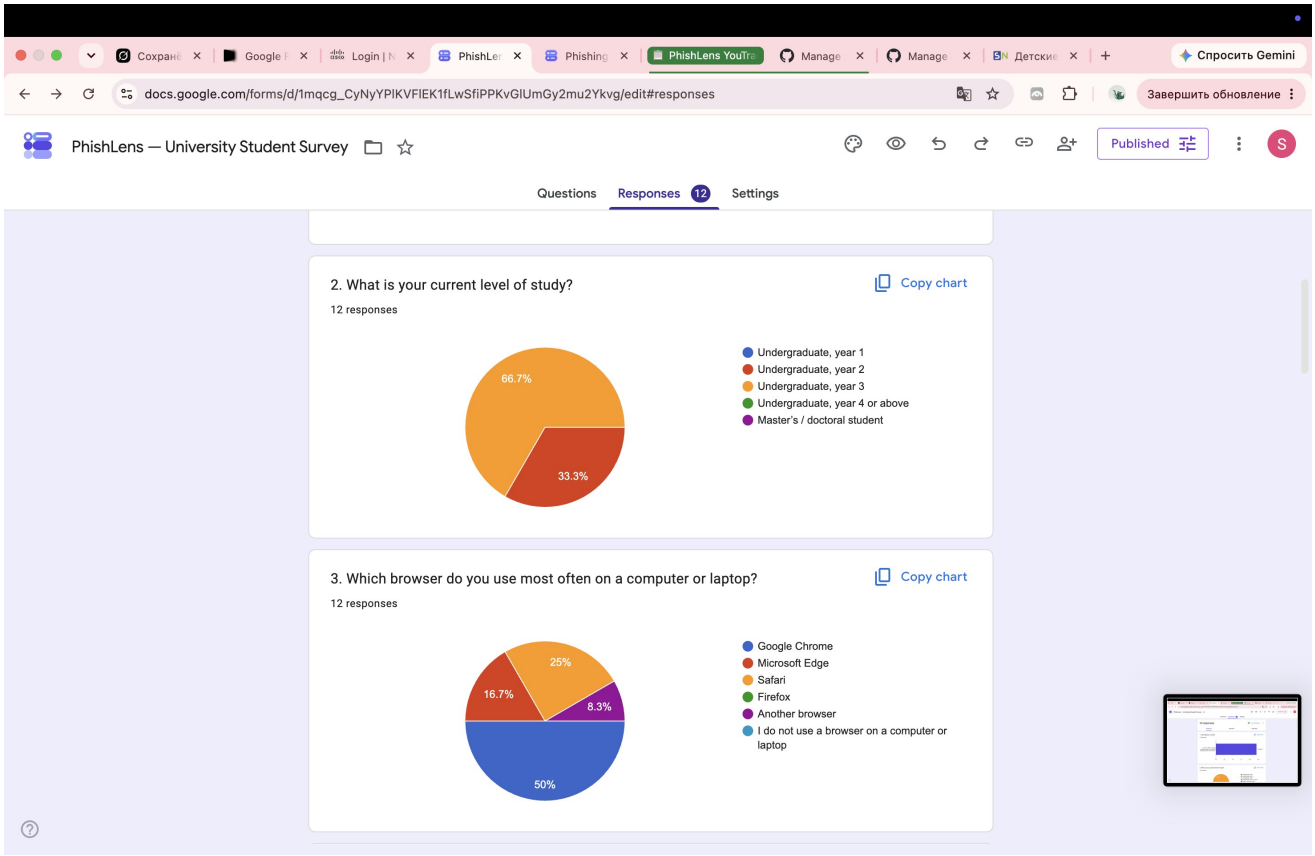
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 1 of 7; captured 24 October 2026.

Appendix A.2 — Survey evidence

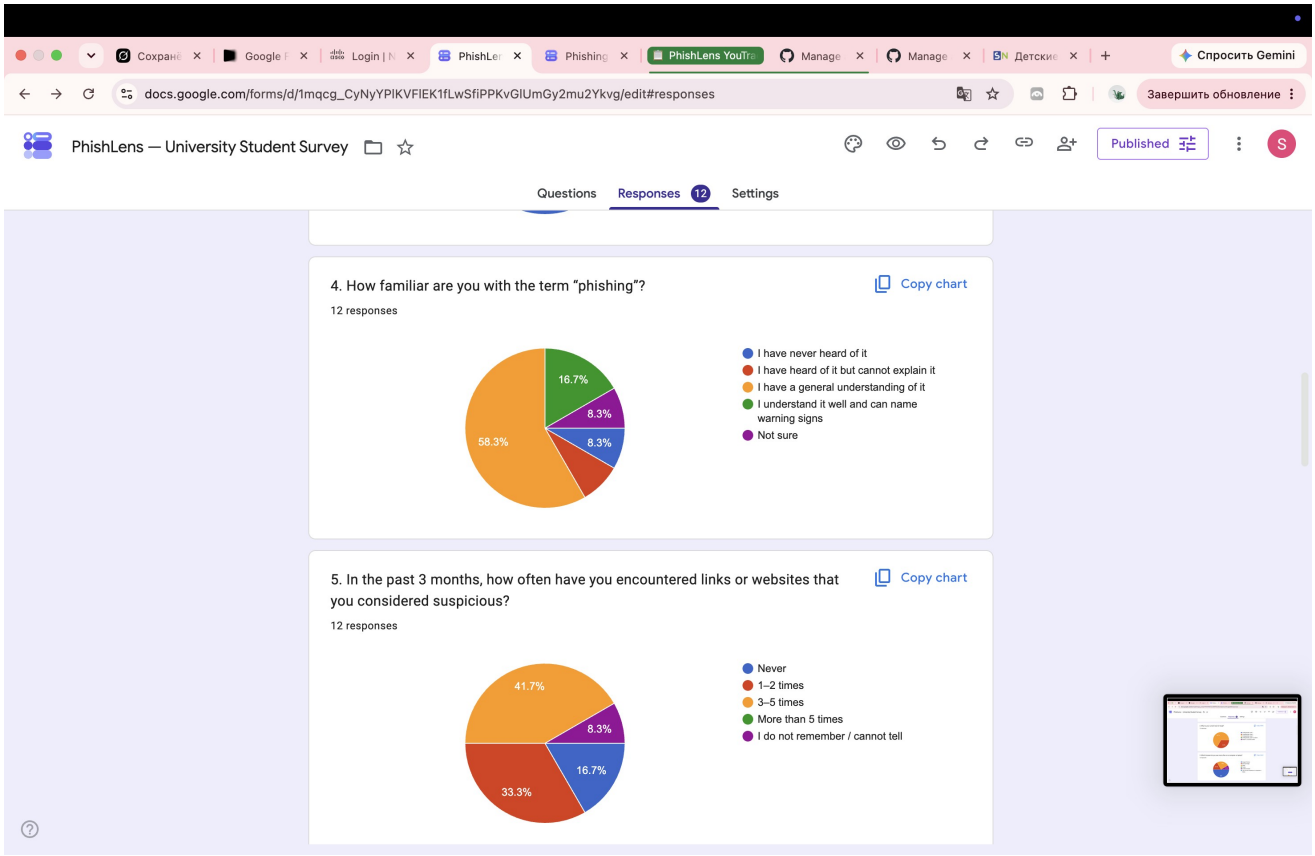
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 2 of 7; captured 24 October 2026.

Appendix A.3 — Survey evidence

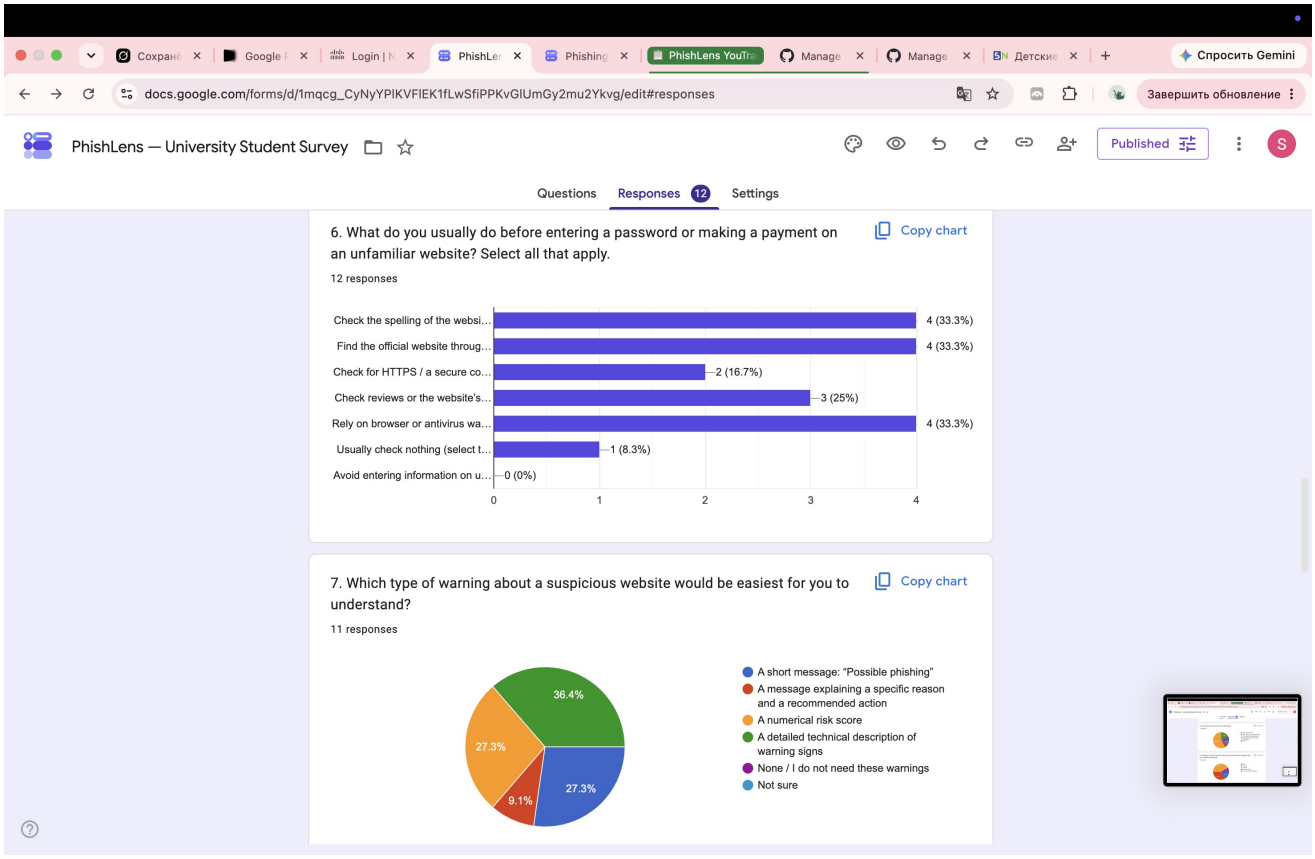
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 3 of 7; captured 24 October 2026.

Appendix A.4 — Survey evidence

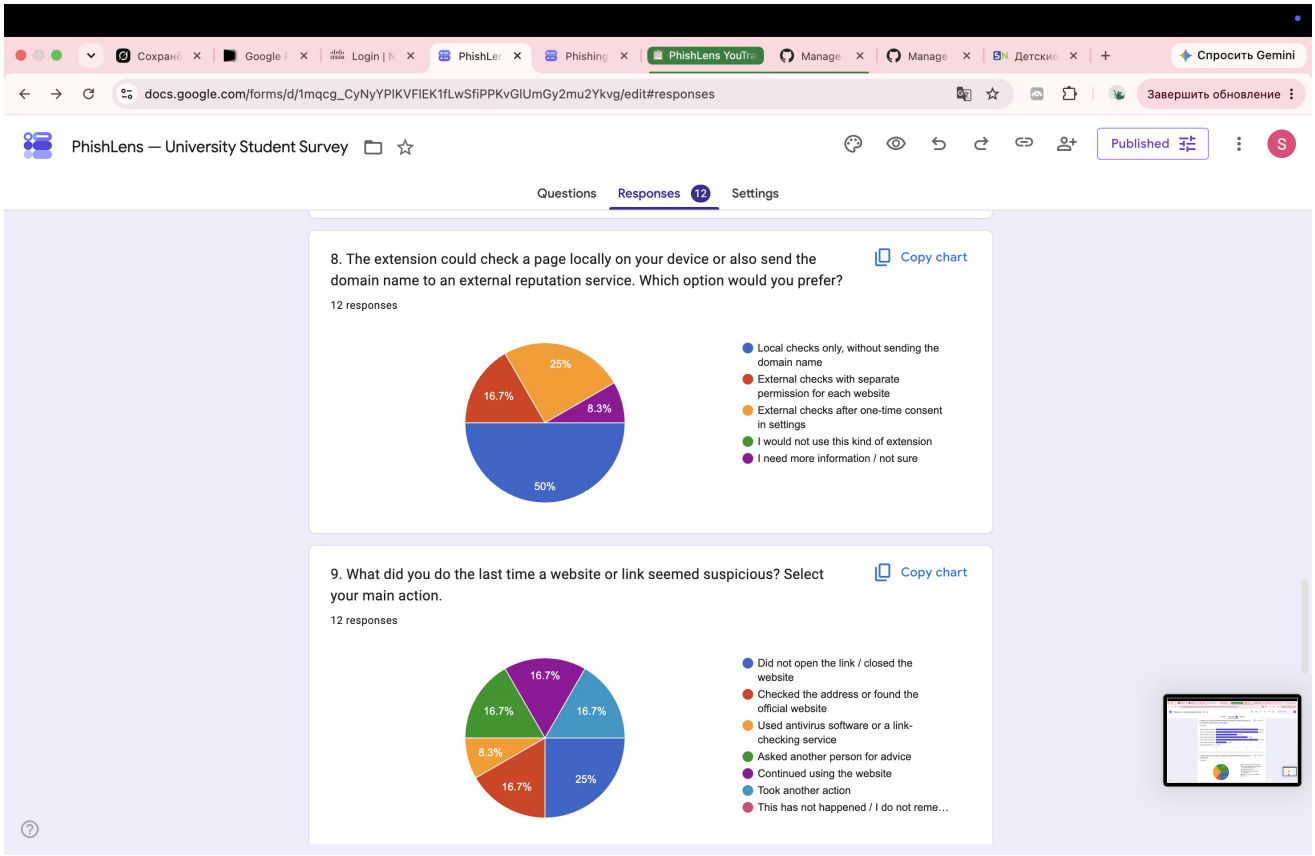
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 4 of 7; captured 24 October 2026.

Appendix A.5 — Survey evidence

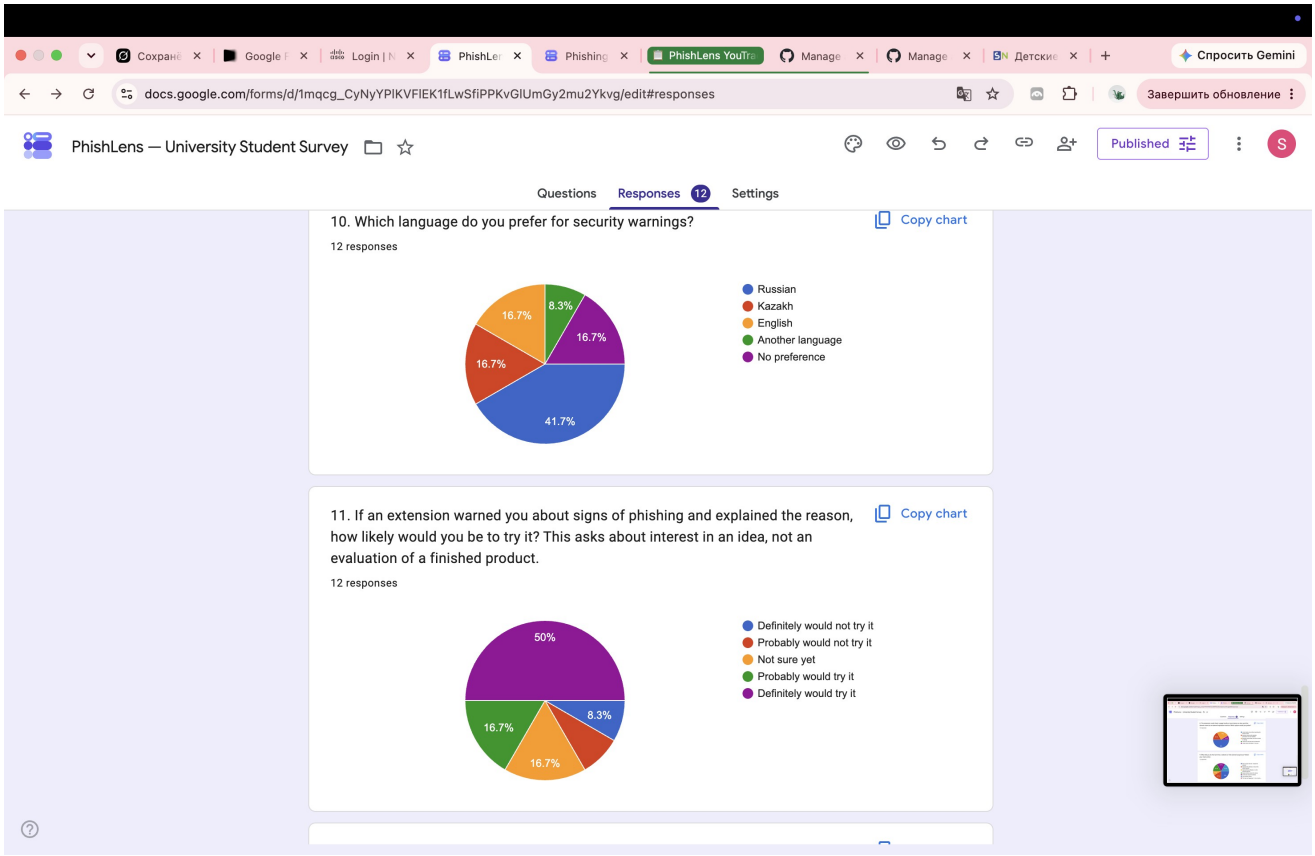
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 5 of 7; captured 24 October 2026.

Appendix A.6 — Survey evidence

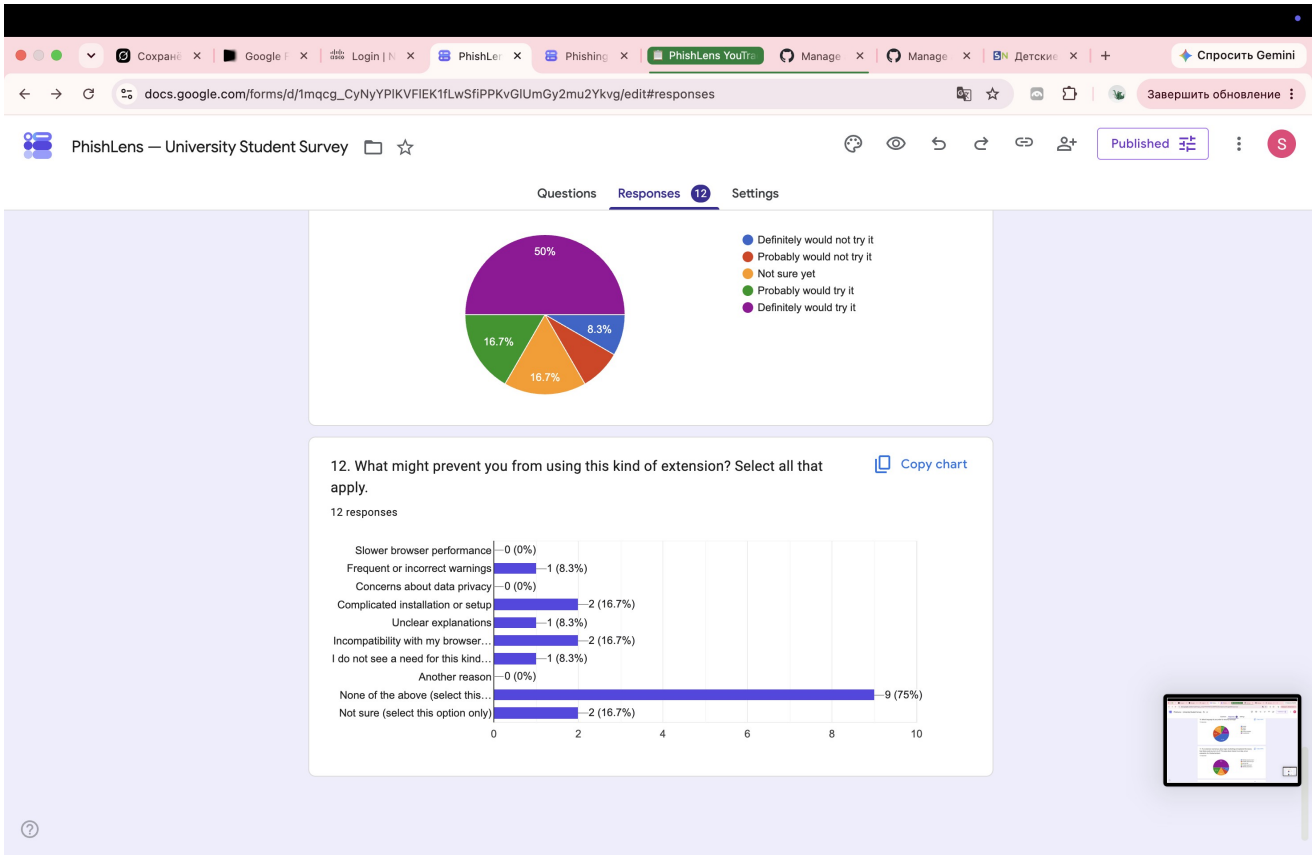
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 6 of 7; captured 24 October 2026.

Appendix A.7 — Survey evidence

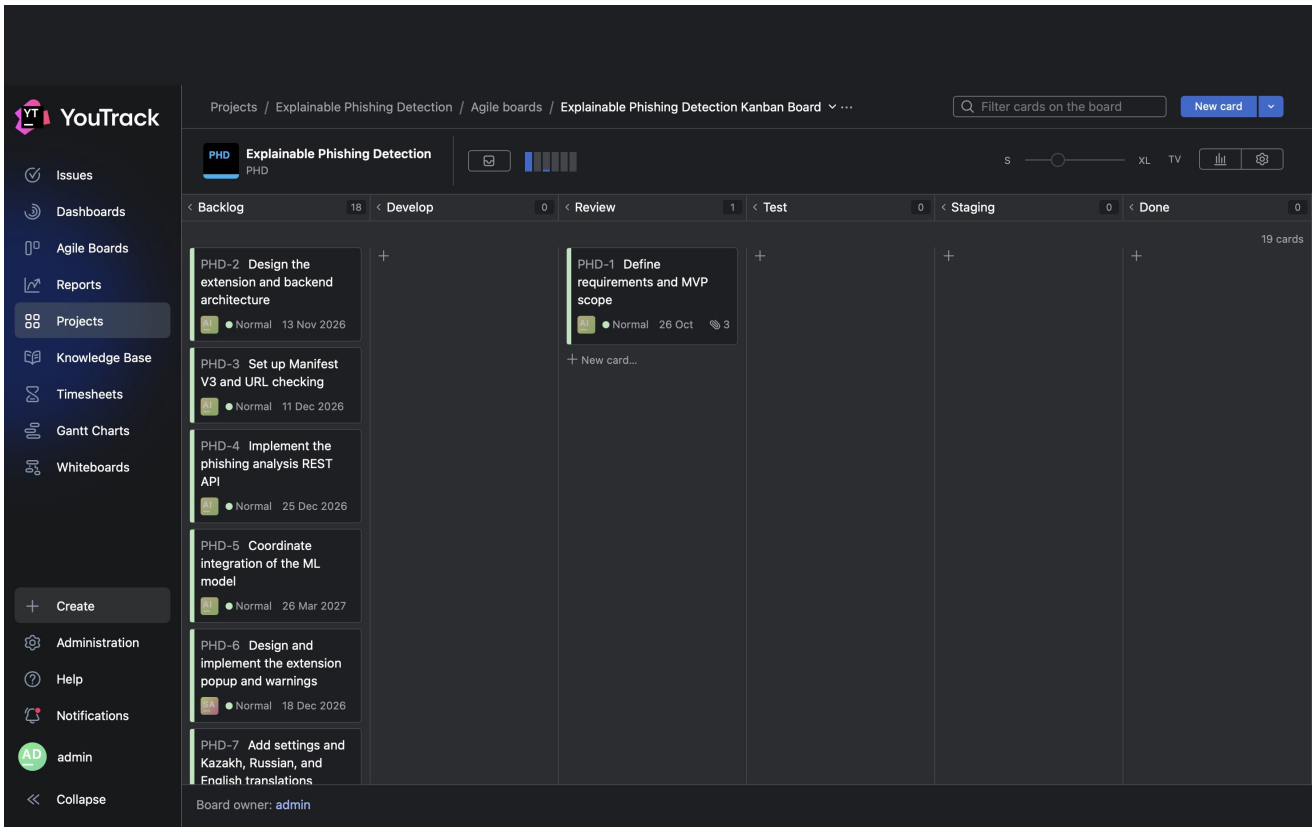
Original Google Forms summary screenshot supplied by the team. Screenshot evidence is retained unchanged; analysis appears in section 2.



Survey summary 7 of 7; captured 24 October 2026.

Appendix B.1 — YouTrack evidence

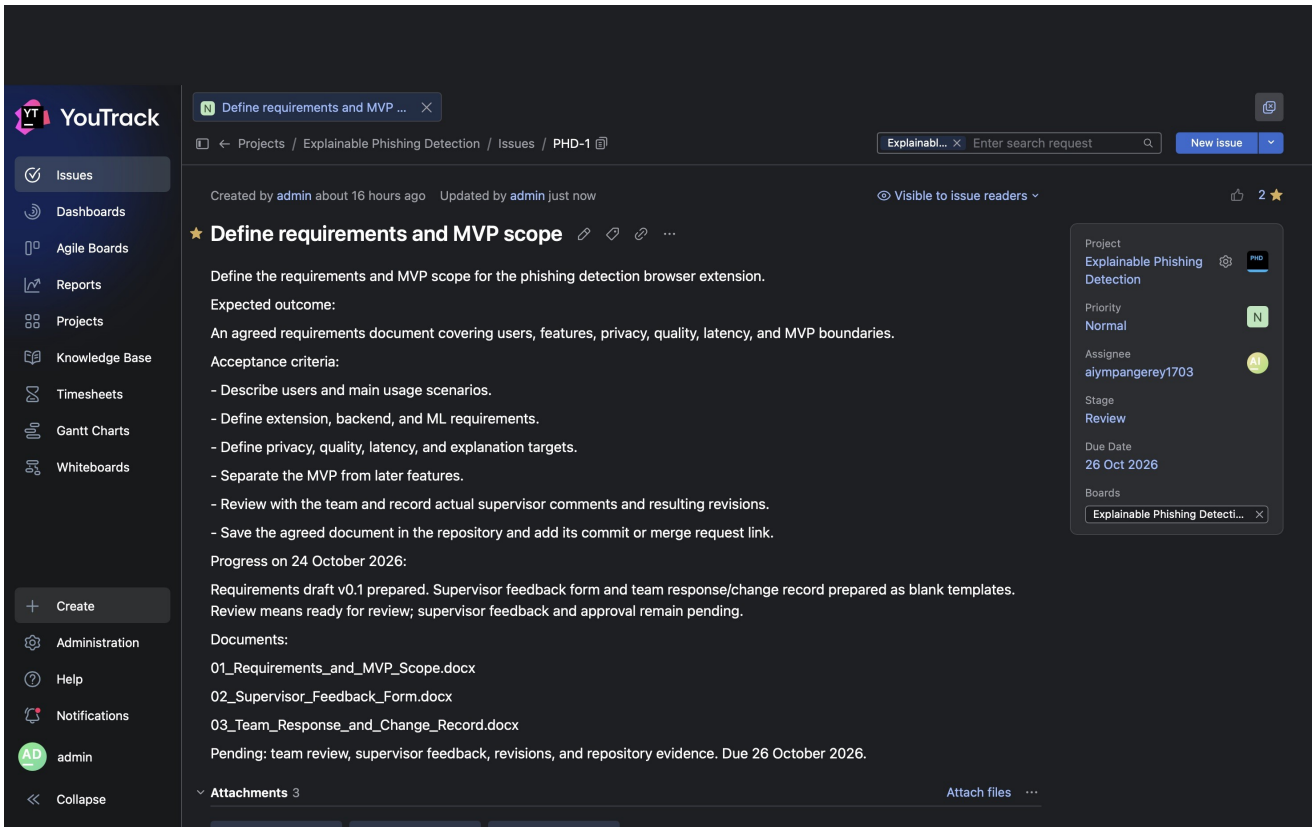
Original screenshot supplied in the review package. A separate landscape PDF provides a larger reading view.



Kanban board snapshot; 18 Backlog and 1 Review.

Appendix B.2 — YouTrack evidence

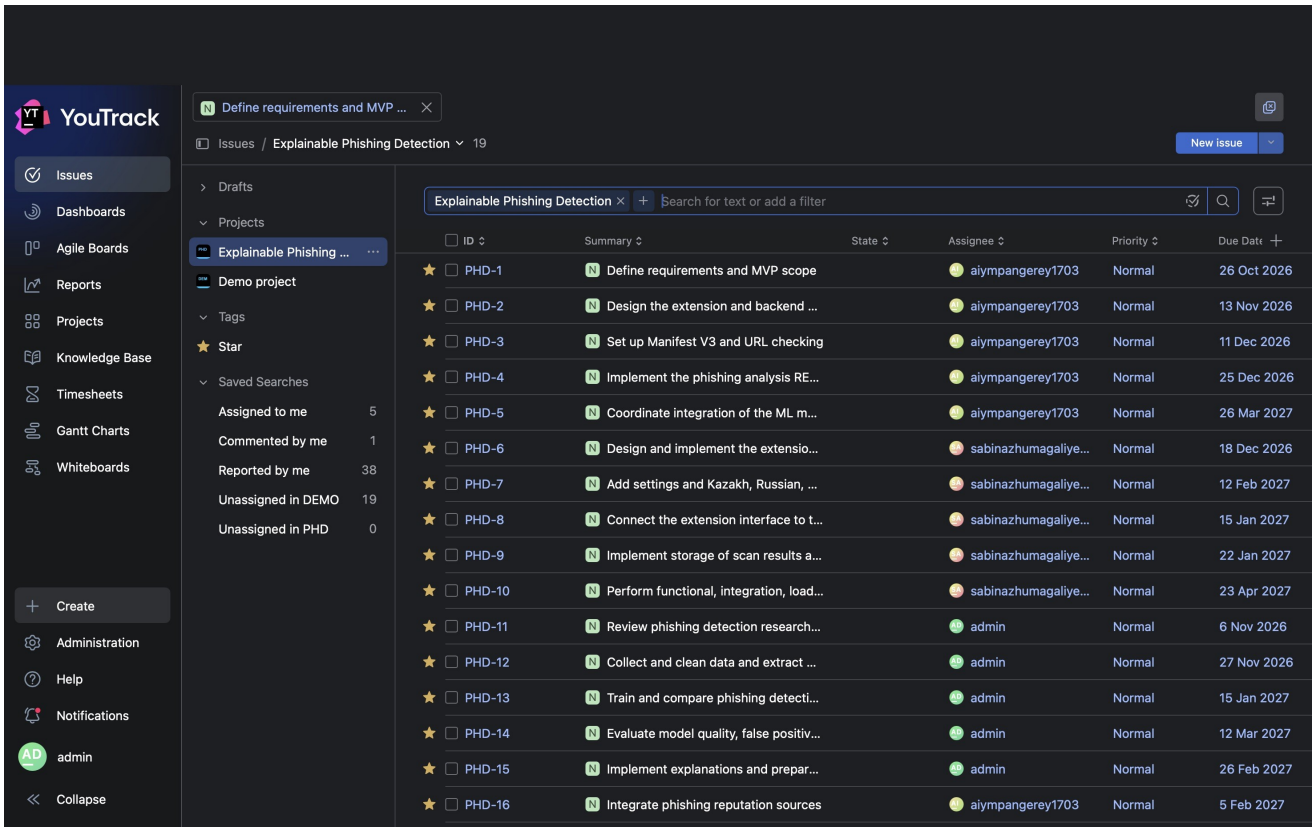
Original screenshot supplied in the review package. A separate landscape PDF provides a larger reading view.



PHD-1 review task with document attachments.

Appendix B.3 — YouTrack evidence

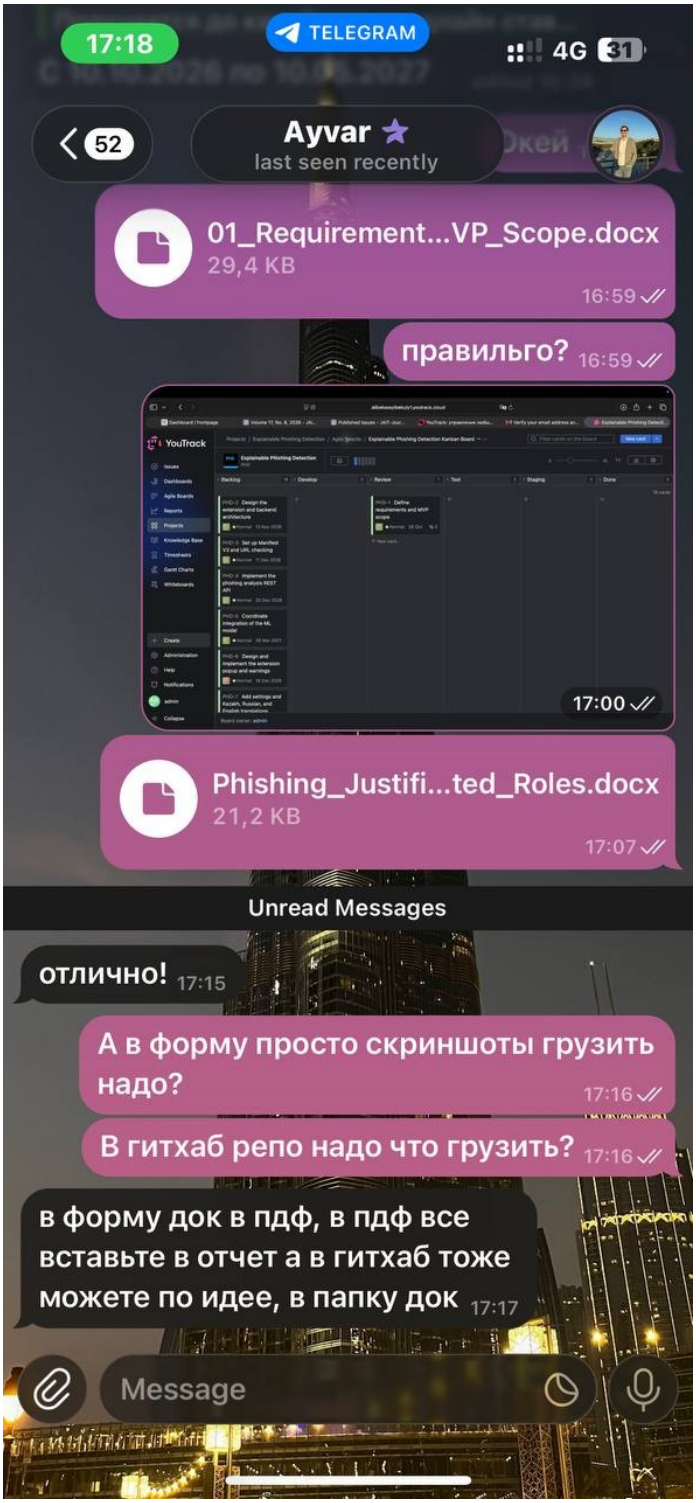
Original screenshot supplied in the review package. A separate landscape PDF provides a larger reading view.



Task list and planned dates.

Appendix C — Supervisor message

Original message screenshot supplied by the team; English interpretation and limits appear in section 11. Russian source text is retained as authentic evidence.



24 October 2026: general positive feedback and PDF packaging guidance.